Cloud Computing

CLOUD COMPUTING PROJECT

PROJECT DETAILS			
Student Name::	**Hieu Le**		
Student ID:	**23201637**		
Project Title:	**US Heart Disease Data Warehouse**	CC Project:	**1**
Due Date:	**06/12/2024**	Submitted Date:	**06/12/2024**

INSTRUCTIONS

1. **Project:** You are asked to choose between two projects. Once you have done that, you need to:
    a. Develop a cloud-based solution for either manage or analyse medical datasets, depending on the project you chose (1 or 2).
    b. Please follow the instructions as described in the project.
2. **Project Report:** produce a project report describing the data set application being studied. This report should contain the following items:
    a. Application overview.
    b. Objectives: Briefly describe each goal/objective and whether they have been completed.
    c. Describe the problem and the collection of the datasets.
    d. Explain your methodology and implementation.
    e. Discuss the suitability of the tool being used to solve the problem.
    f. Describe the features of your software.
    g. Give a worked example.
    h. Conclusion
    i. References
3. **Submission:** The deadline is 06 December 2024. Passed this deadline penalty will apply.
    a. Please submit your project report, the source code, and all the necessary files to execute your implementation.
    b. You may need to prepare a README file to explain how to execute it.
    c. Please submit ONLY one ZIP file, containing all the necessary files and directories, including the project report.

Application Overview

"US Heart Disease Dashboard" combines data warehouse, data analytics, and data visualization to help healthcare organizations, medical researchers, and public health officials understand and address cardiovascular disease risks across the United States. Its ultimate goal is to drive data-driven cardiovascular health program decision-making by utilizing multidimensional health data in a user-friendly format. The dashboard leverages Google Cloud BigQuery’s data warehouse capabilities to process large-scale health survey responses while providing Looker Studio’s interactive visualizations to offer actionable insights via three components: Demographic analysis, lifestyle impact assessment, and preventive care evaluation. These components support the three critical use cases as follows:

1. Public health administrators and policymakers can analyze demographic patterns and regional variations in heart disease prevalence to allocate resources and design targeted intervention programs. The geographic visualization and demographic breakdowns help identify underserved populations and areas requiring additional healthcare support.

2. Medical researchers can investigate the relationships between lifestyle factors, such as smoking, physical activity, sleep patterns, and heart attack rates. The interactive charts enable the exploration of multiple variables and their correlations with cardiovascular health outcomes.

3. Healthcare providers can evaluate the effectiveness of preventive care measures by examining how regular checkups, vaccinations, and other preventive services correlate with heart attack rates across different population groups. These insights help develop and refine public health recommendations and outreach programs.

Objectives

The project’s primary goal is to leverage cloud-based data analytics and close the gap between multidimensional health data and practical insights for cardiovascular health management. This can be broken down into two objectives: The first objective is to create a data analytics platform that processes and analyzes the nationwide-level health survey data through Google BigQuery. This involves designing an efficient data warehouse structure that can handle complex health datasets while maintaining data quality and accessibility. This objective has been accomplished as the platform can process information from over 246,000 survey participants.

The second objective is to develop interactive visualization dashboards that transform complex health data into actionable insights. This involves creating three specialized dashboard views with cross-filtering functionalities: a demographic analysis dashboard showing population-level patterns, a lifestyle impact dashboard examining behavioral risk factors, and a preventive care dashboard tracking intervention methods’ effectiveness. Within each dashboard, interaction capabilities allow different user groups to explore and analyze data relevant to their needs. All components of this objective have also been completed with each interactive dashboard successfully implemented and functional.

Problem Statement & Data Collection

U.S. healthcare organizations fight a war against heart disease - the leading cause of death for most racial and ethnic groups, despite having access to extensive health survey data (Centers for Disease Control and Prevention [CDC], 2024). One of the core problems lies in transforming complex health information into actionable insights across three key areas: understanding demographic risk patterns for resource allocation, quantifying the impact of lifestyle choices on heart health, and evaluating the effectiveness of preventive care measures (Khedo, K.K. et al., 2020). Traditional analysis methods struggle to process the multidimensional nature of cardiovascular health data, making it difficult for public healthcare administrators, clinical researchers, and health providers to make data-driven decisions. This creates a pressing need for a comprehensive analytics platform that can process large-scale survey data while presenting insights in a user-friendly yet actionable format for different stakeholders.

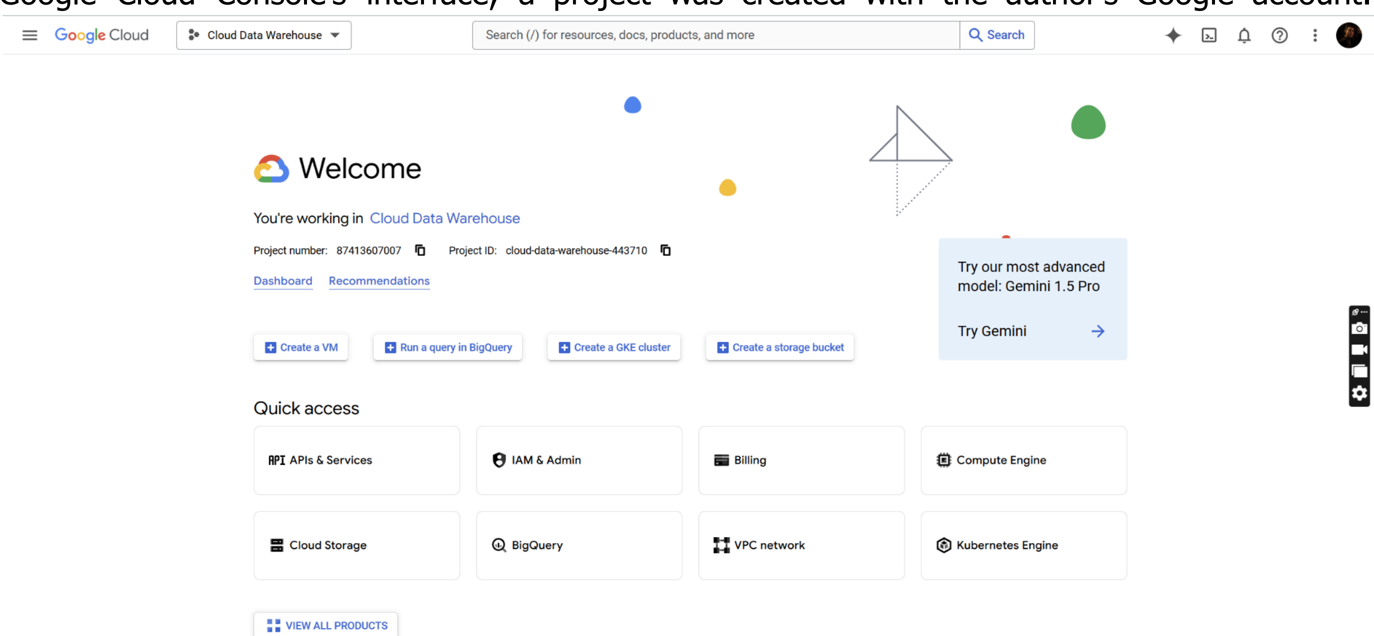
Regarding data collection, the dashboard uses the Kaggle dataset "Indicators of Heart Disease", which was processed and sourced from the Behavioral Risk Factor Surveillance System (BRFSS), the world's largest continuous health survey conducted by the CDC. This dataset includes responses from over 246,000 adults and provides insights into personal health behaviors, medical history, and preventive care practices. While there are 40 variables related to heart disease, only a subset of these variables is used in the dashboard design to accommodate the aforementioned use cases (as highlighted below). The full variable list with description, which has been modified based on the dataset author's GitHub, is as follows:

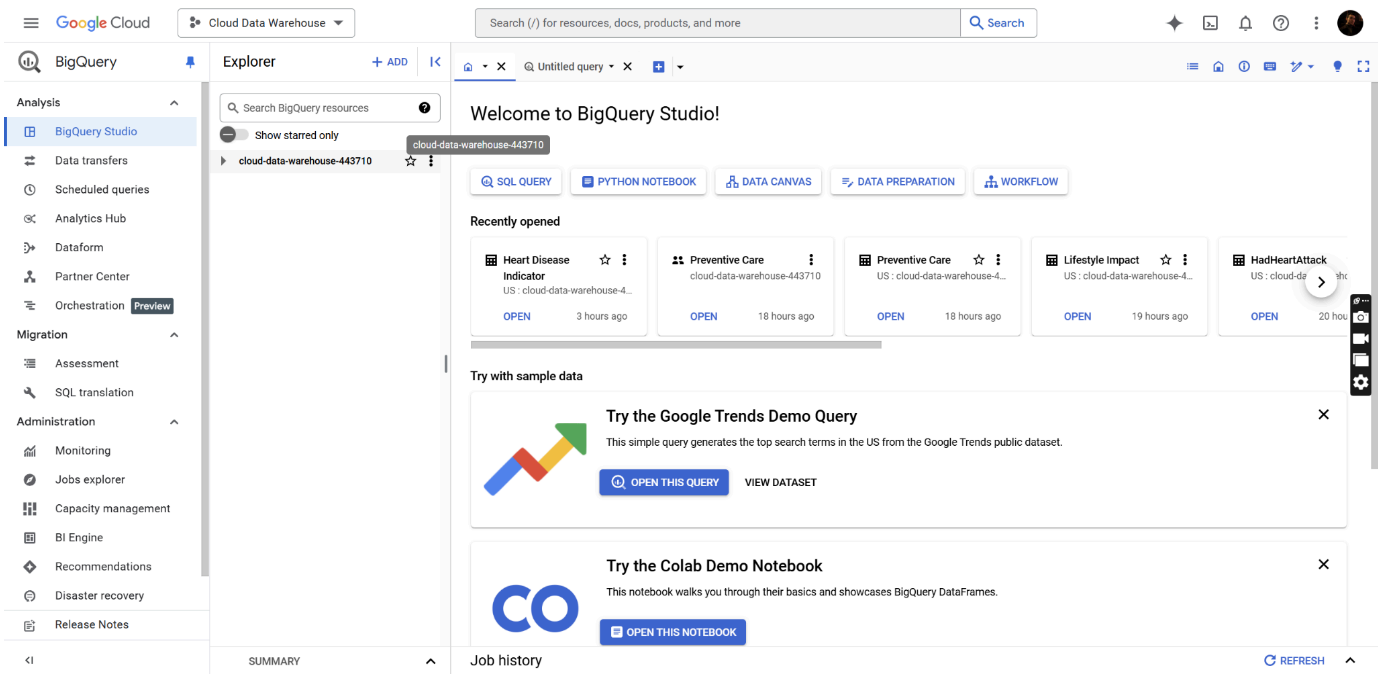
1. **State**: State FIPS Code.
2. **Sex**: Sex of the respondent.
3. GeneralHealth: General health status (e.g., excellent, good, fair, poor).
4. **PhysicalHealthDays**: Number of days in the past month when physical health was not good.
5. **MentalHealthDays**: Number of days in the past month when mental health was not good.
6. **LastCheckupTime**: Time since the last routine checkup.
7. **PhysicalActivities**: Participation in physical activities or exercises in the past month.
8. **SleepHours**: Average number of sleep hours in a 24-hour period.
9. RemovedTeeth: Number of permanent teeth removed due to decay or gum disease.
10. **HadHeartAttack**: Whether ever told they had a heart attack.
11. HadAngina: Whether ever told they had angina or coronary heart disease.
12. HadStroke: Whether ever told they had a stroke.
13. HadAsthma: Whether ever told they had asthma.
14. HadSkinCancer: Whether ever told they had skin cancer.
15. HadCOPD: Whether ever told they had COPD, emphysema, or chronic bronchitis.
16. HadDepressiveDisorder: Whether ever told they had a depressive disorder.
17. HadKidneyDisease: Whether ever told they had kidney disease.
18. HadArthritis: Whether ever told they had arthritis.
19. HadDiabetes: Whether ever told they had diabetes.
20. DeafOrHardOfHearing: Whether they are deaf or have serious difficulty hearing.
21. BlindOrVisionDifficulty: Whether they are blind or have serious difficulty seeing.

22. DifficultyConcentrating: Serious difficulty concentrating, remembering, or making decisions due to a condition.
23. DifficultyWalking: Serious difficulty walking or climbing stairs.
24. DifficultyDressingBathing: Difficulty dressing or bathing.
25. DifficultyErrands: Difficulty doing errands alone due to a physical, mental, or emotional condition.
26. **SmokerStatus**: Smoker status (e.g., everyday smoker, former smoker, non-smoker).
27. ECigaretteUsage: E-cigarette usage status (e.g., never used, current user, past user).
28. ChestScan: Whether ever had a CT or CAT scan of the chest area.
29. **RaceEthnicityCategory**: Five-level race/ethnicity category.
30. **AgeCategory**: Fourteen-level age category.
31. HeightInMeters: Reported height in meters.
32. WeightInKilograms: Reported weight in kilograms.
33. **BMI**: Body Mass Index (BMI).
34. **AlcoholDrinkers**: Whether had at least one drink of alcohol in the past 30 days.
35. HIVTesting: Whether ever tested for HIV.
36. **FluVaxLast12**: Whether had a flu vaccine in the past 12 months.
37. **PneumoVaxEver**: Whether ever had a pneumococcal vaccine.
38. TetanusLast10Tdap: Whether had a tetanus shot in the past 10 years (specifically Tdap).
39. HighRiskLastYear: Whether engaged in high-risk activities (e.g., drug injection, STD treatment, exchanged sex for money/drugs) in the past year.
40. CovidPos: Whether ever tested positive for COVID-19.

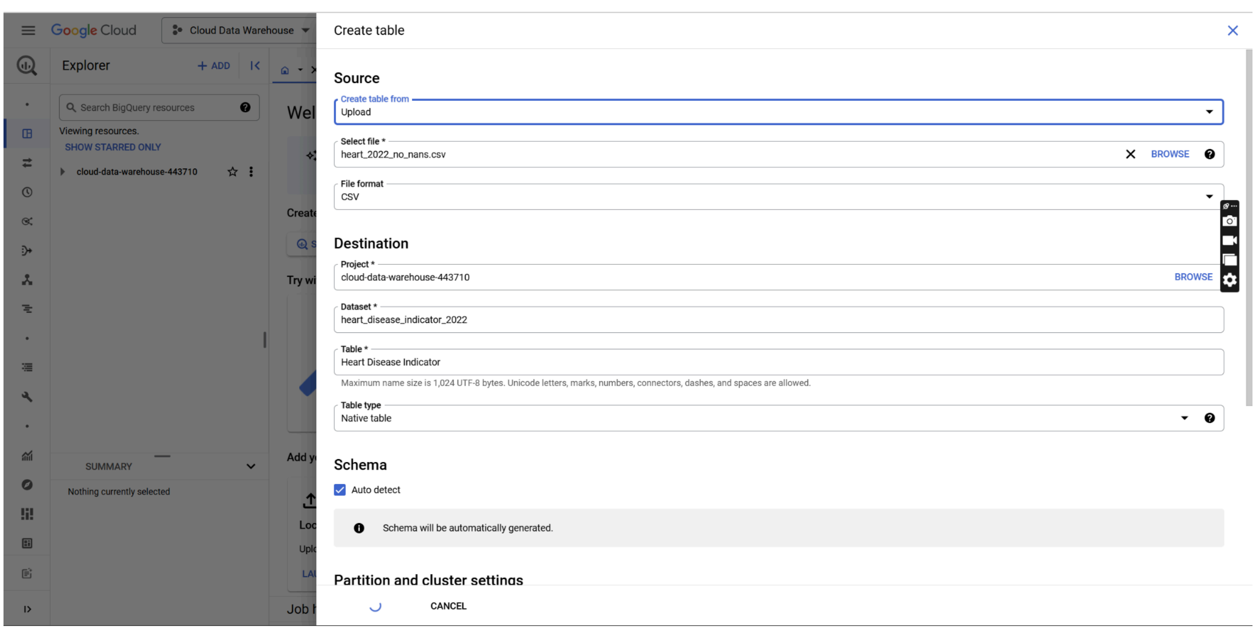
Methodology & Implementation

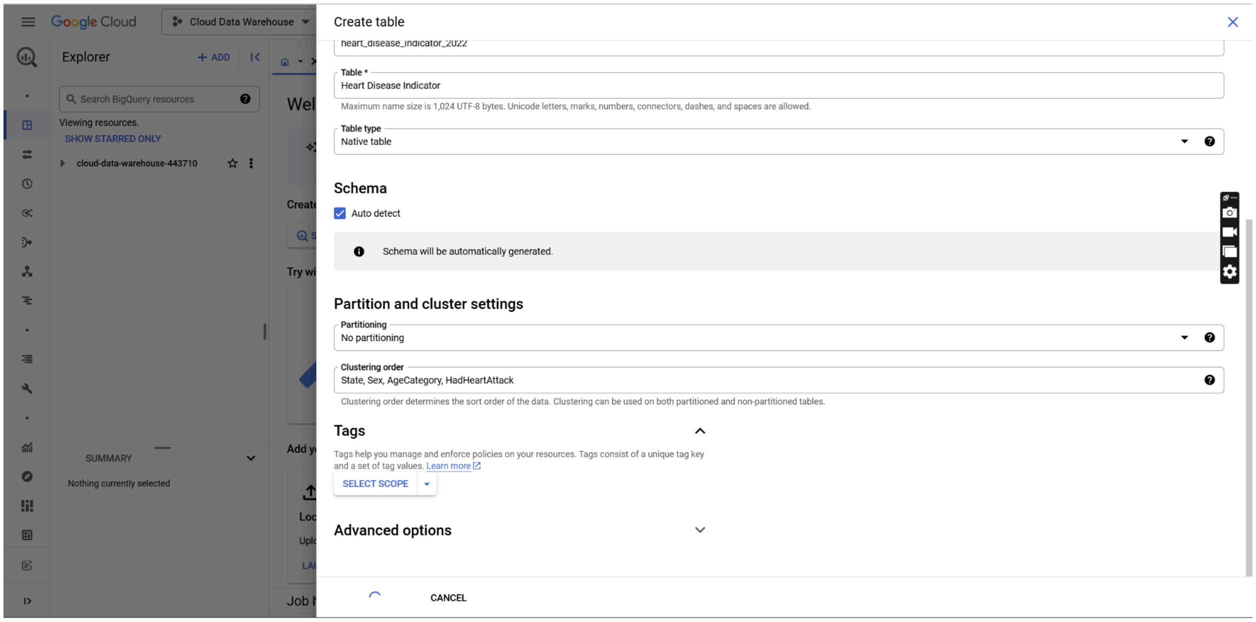
The tool leverages Google Cloud Platform's integrated services with BigQuery handling data warehousing and Looker Studio providing visualization capabilities. To start with, from the Google Cloud Console's interface, a project was created with the author's Google account.



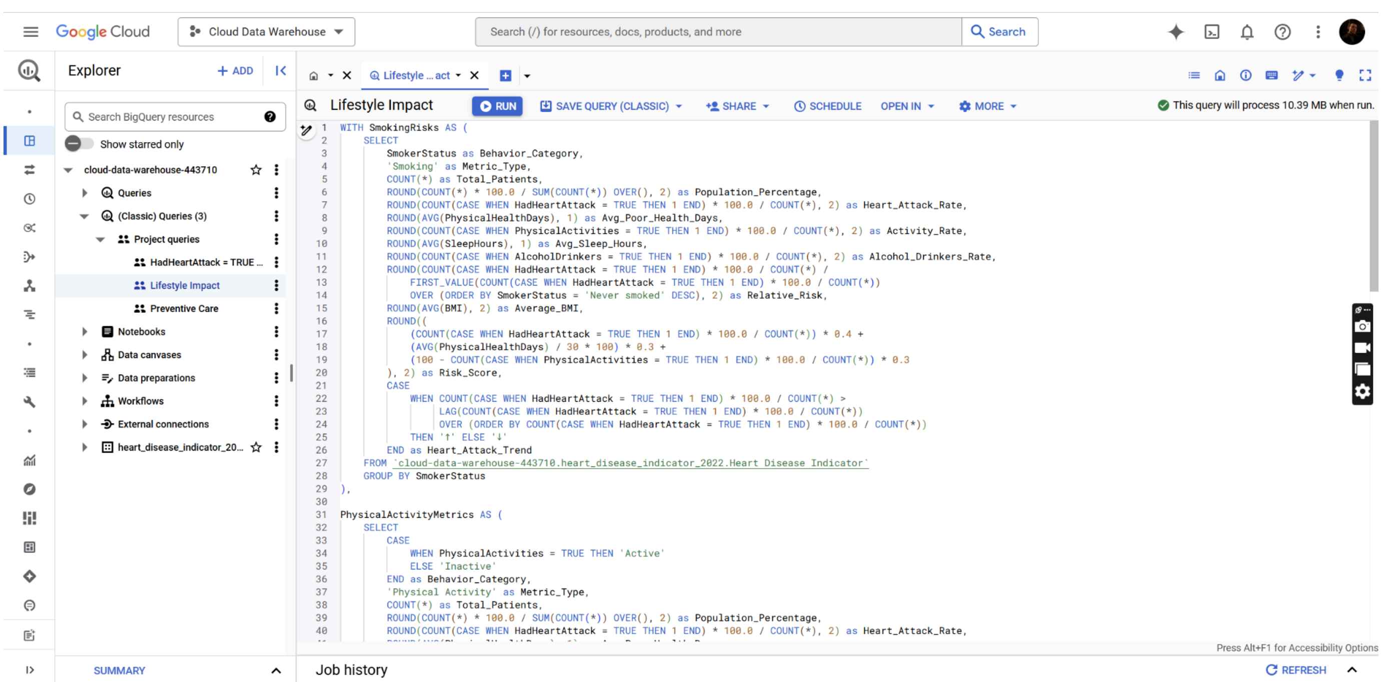


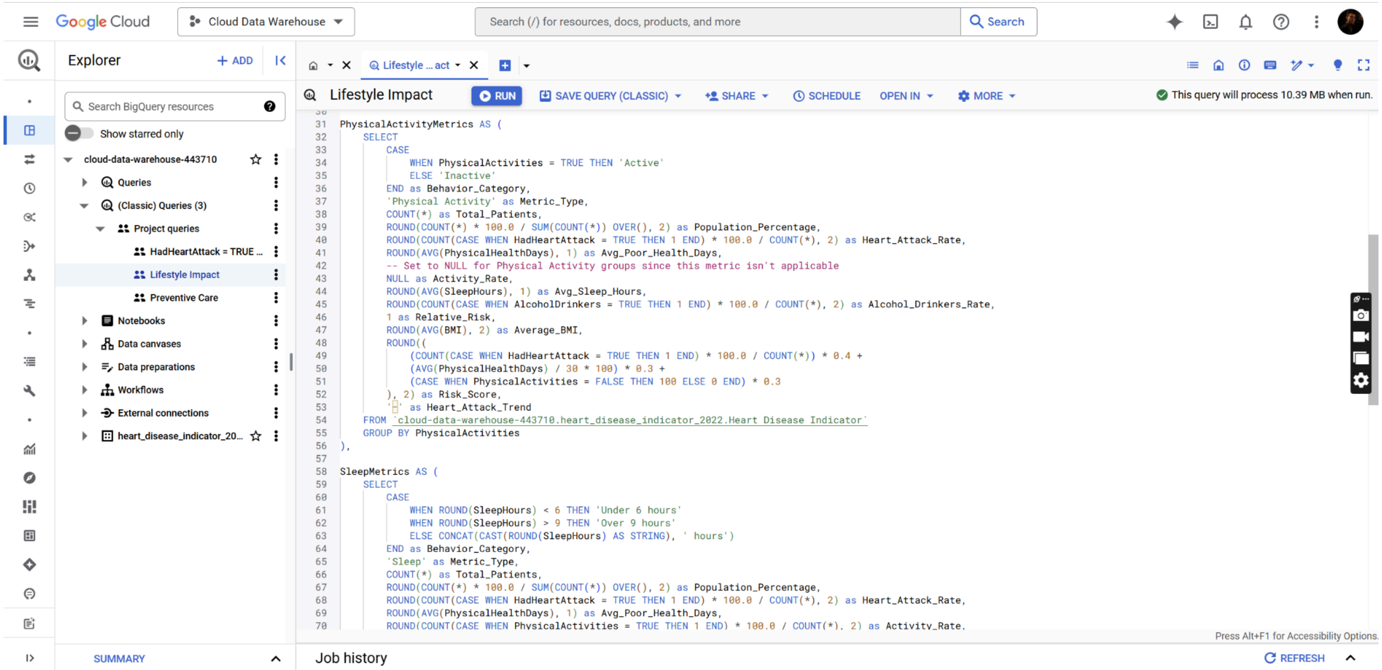
Then, the implementation for data processing was done with BigQuery's serverless architecture, which does not require infrastructure setup or server management. Without the initial setup, the Kaggle dataset flew through two stages: data ingestion and data transformation. In the data ingestion stage, the dataset was uploaded locally from the host computer. Clustering order was applied for certain variables that were frequently queried when creating the first dashboard. It is noteworthy that BigQuery can also ingest data through both batch processing and streaming ingestion for real-time updates, which do not belong to the scope of this project. Nonetheless, if this was for real-life use case, this dual approach would help healthcare organizations analyze both long-term cardiovascular health trends while maintaining current insights.





In the data transformation stage, as the data entered the warehouse, it underwent several transformations. The author first wrote SQL queries to select key relevant variables such as demographics, lifestyle behaviors, and preventive care metrics, as previously noted from the Data Collection section, for constructing tables. Then, the author proceeded to standardize the format of health metrics, grouped related metrics together, calculate derived risk factors as well as other relevant metrics such as heart attack rates or average BMI, and create aggregated views for different demographic segments. These transformations prepared the data for efficient analysis by the tool's users in the subsequent visualization step.




Google Cloud
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cloud-data-warehouse-443710
Queries
(Classic) Queries (3)
Project queries
HadHeartAttack = TRUE ...
Lifestyle Impact
Preventive Care
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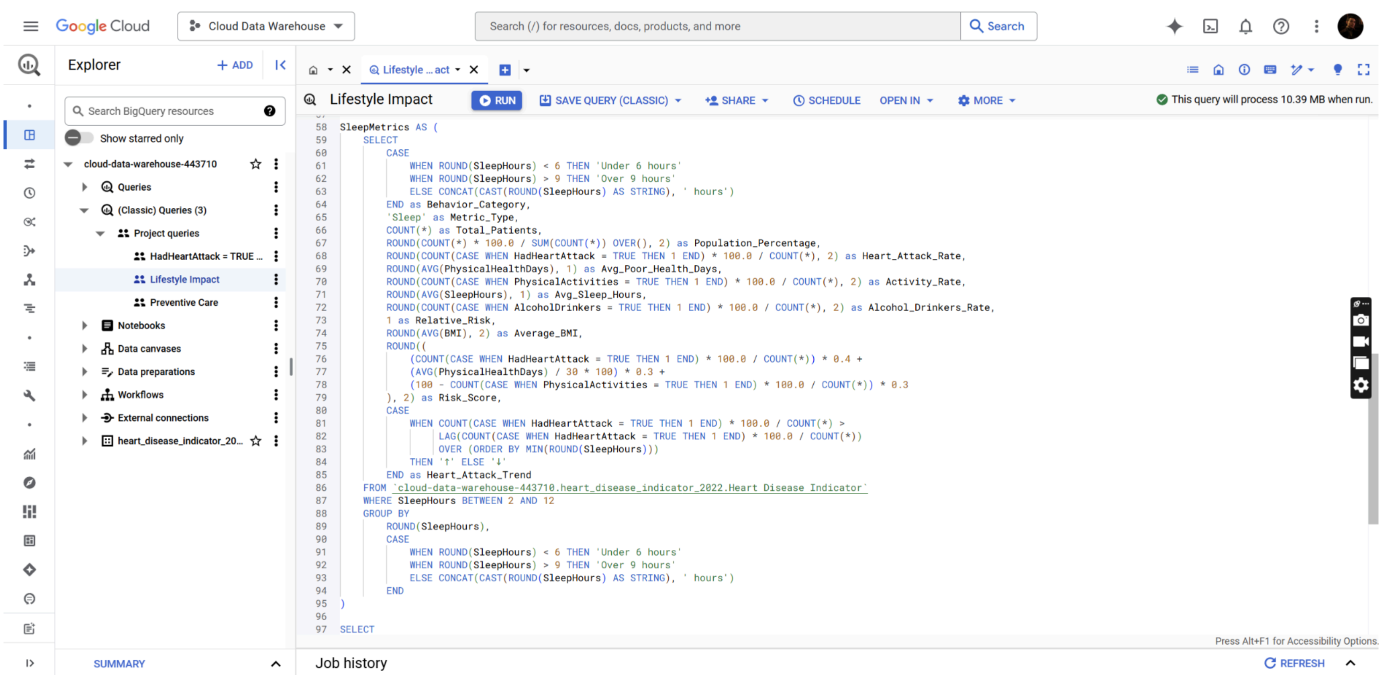
Data canvases
Data preparations
Workflows
External connections
heart_disease_indicator_20...
SUMMARY
Lifestyle ... act
Lifestyle Impact
RUN
SAVE QUERY (CLASSIC)
SHARE
SCHEDULE
OPEN IN
MORE
This query will process 10.39 MB when run.
PhysicalActivityMetrics AS (
SELECT
CASE
WHEN PhysicalActivities = TRUE THEN 'Active'
ELSE 'Inactive'
END as Behavior_Category,
'Physical Activity' as Metric_Type,
COUNT(*) as Total_Patients,
ROUND(COUNT(*) * 100.0 / SUM(COUNT(*)) OVER(), 2) as Population_Percentage,
ROUND(COUNT(CASE WHEN HadHeartAttack = TRUE THEN 1 END) * 100.0 / COUNT(*), 2) as Heart_Attack_Rate,
ROUND(AVG(PhysicalHealthDays), 1) as Avg_Poor_Health_Days,
-- Set to NULL for Physical Activity groups since this metric isn't applicable
NULL as Activity_Rate,
ROUND(AVG(SleepHours), 1) as Avg_Sleep_Hours,
ROUND(COUNT(CASE WHEN AlcoholDrinkers = TRUE THEN 1 END) * 100.0 / COUNT(*), 2) as Alcohol_Drinkers_Rate,
1 as Relative_Risk,
ROUND(AVG(BMI), 2) as Average_BMI,
ROUND((
(COUNT(CASE WHEN HadHeartAttack = TRUE THEN 1 END) * 100.0 / COUNT(*)) * 0.4 +
(AVG(PhysicalHealthDays) / 30 * 100) * 0.3 +
(CASE WHEN PhysicalActivities = FALSE THEN 100 ELSE 0 END) * 0.3
), 2) as Risk_Score,
'-' as Heart_Attack_Trend
FROM `cloud-data-warehouse-443710.heart_disease_indicator_2022.Heart Disease Indicator`
GROUP BY PhysicalActivities
),
SleepMetrics AS (
SELECT
CASE
WHEN ROUND(SleepHours) < 6 THEN 'Under 6 hours'
WHEN ROUND(SleepHours) > 9 THEN 'Over 9 hours'
ELSE CONCAT(CAST(ROUND(SleepHours) AS STRING), ' hours')
END as Behavior_Category,
'Sleep' as Metric_Type,
COUNT(*) as Total_Patients,
ROUND(COUNT(*) * 100.0 / SUM(COUNT(*)) OVER(), 2) as Population_Percentage,
ROUND(COUNT(CASE WHEN HadHeartAttack = TRUE THEN 1 END) * 100.0 / COUNT(*), 2) as Heart_Attack_Rate,
ROUND(AVG(PhysicalHealthDays), 1) as Avg_Poor_Health_Days,
ROUND(COUNT(CASE WHEN PhysicalActivities = TRUE THEN 1 END) * 100.0 / COUNT(*), 2) as Activity_Rate,
Press Alt+F1 for Accessibility Options.
Job history
REFRESH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END as Behavior_Category,
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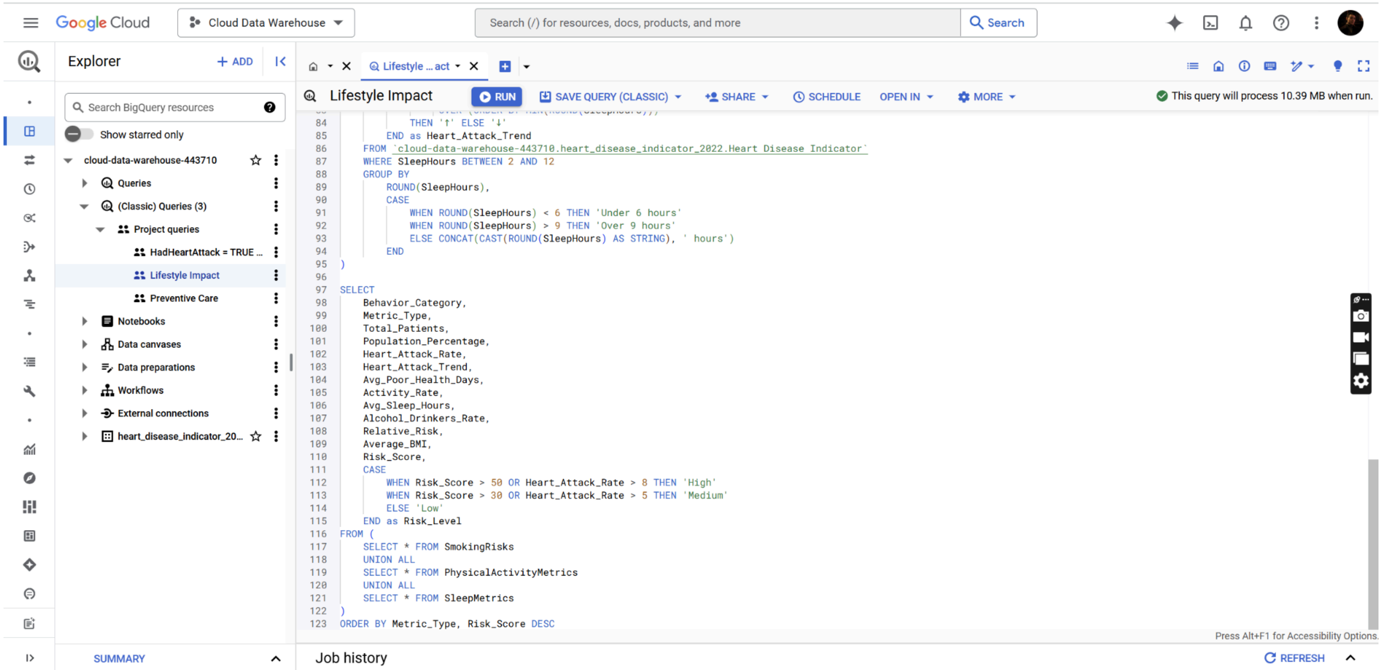
COUNT(*) as Total_Patients,
ROUND(COUNT(*) * 100.0 / SUM(COUNT(*)) OVER(), 2) as Population_Percentage,
ROUND(COUNT(CASE WHEN HadHeartAttack = TRUE THEN 1 END) * 100.0 / COUNT(*), 2) as Heart_Attack_Rate,
ROUND(AVG(PhysicalHealthDays), 1) as Avg_Poor_Health_Days,
ROUND(COUNT(CASE WHEN PhysicalActivities = TRUE THEN 1 END) * 100.0 / COUNT(*), 2) as Activity_Rate,
ROUND(AVG(SleepHours), 1) as Avg_Sleep_Hours,
ROUND(COUNT(CASE WHEN AlcoholDrinkers = TRUE THEN 1 END) * 100.0 / COUNT(*), 2) as Alcohol_Drinkers_Rate,
1 as Relative_Risk,
ROUND(AVG(BMI), 2) as Average_BMI,
ROUND((
(COUNT(CASE WHEN HadHeartAttack = TRUE THEN 1 END) * 100.0 / COUNT(*)) * 0.4 +
(AVG(PhysicalHealthDays) / 30 * 100) * 0.3 +
(100 - COUNT(CASE WHEN PhysicalActivities = TRUE THEN 1 END) * 100.0 / COUNT(*)) * 0.3
), 2) as Risk_Score,
CASE
WHEN COUNT(CASE WHEN HadHeartAttack = TRUE THEN 1 END) * 100.0 / COUNT(*) >
LAG(COUNT(CASE WHEN HadHeartAttack = TRUE THEN 1 END) * 100.0 / COUNT(*))
OVER (ORDER BY MIN(ROUND(SleepHours)))
THEN '↑' ELSE '↓'
END as Heart_Attack_Trend
FROM `cloud-data-warehouse-443710.heart_disease_indicator_2022.Heart Disease Indicator`
WHERE SleepHours BETWEEN 2 AND 12
GROUP BY
ROUND(SleepHours),
CASE
WHEN ROUND(SleepHours) < 6 THEN 'Under 6 hours'
WHEN ROUND(SleepHours) > 9 THEN 'Over 9 hours'
ELSE CONCAT(CAST(ROUND(SleepHours) AS STRING), ' hours')
END
)
SELECT
Press Alt+F1 for Accessibility Options.
Job history
REFRESH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END as Heart_Attack_Trend
FROM `cloud-data-warehouse-443710.heart_disease_indicator_2022.Heart Disease Indicator`
WHERE SleepHours BETWEEN 2 AND 12
GROUP BY
ROUND(SleepHours),
CASE
WHEN ROUND(SleepHours) < 6 THEN 'Under 6 hours'
WHEN ROUND(SleepHours) > 9 THEN 'Over 9 hours'
ELSE CONCAT(CAST(ROUND(SleepHours) AS STRING), ' hours')
END
)
SELECT
Behavior_Category,
Metric_Type,
Total_Patients,
Population_Percentage,
Heart_Attack_Rate,
Heart_Attack_Trend,
Avg_Poor_Health_Days,
Activity_Rate,
Avg_Sleep_Hours,
Alcohol_Drinkers_Rate,
Relative_Risk,
Average_BMI,
Risk_Score,
CASE
WHEN Risk_Score > 50 OR Heart_Attack_Rate > 8 THEN 'High'
WHEN Risk_Score > 30 OR Heart_Attack_Rate > 5 THEN 'Medium'
ELSE 'Low'
END as Risk_Level
FROM (
SELECT * FROM SmokingRisks
UNION ALL
SELECT * FROM PhysicalActivityMetrics
UNION ALL
SELECT * FROM SleepMetrics
)
ORDER BY Metric_Type, Risk_Score DESC
Press Alt+F1 for Accessibility Options.
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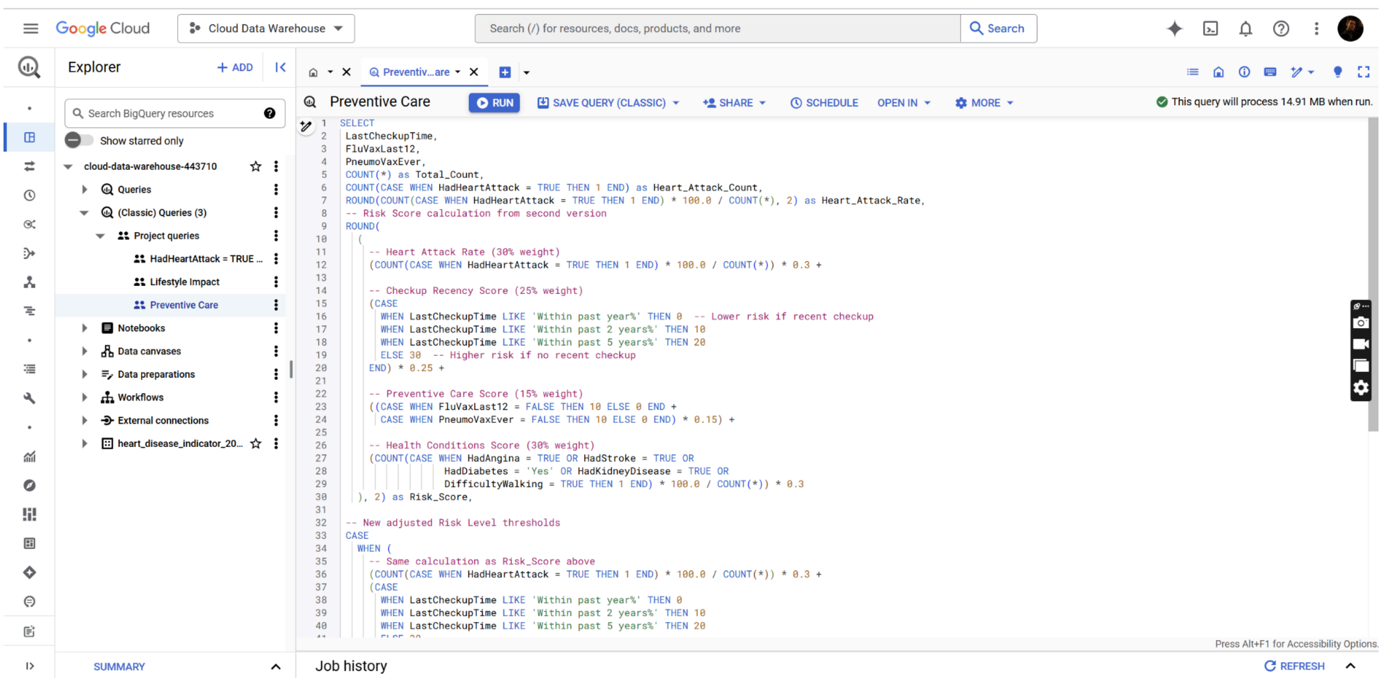
REFRESH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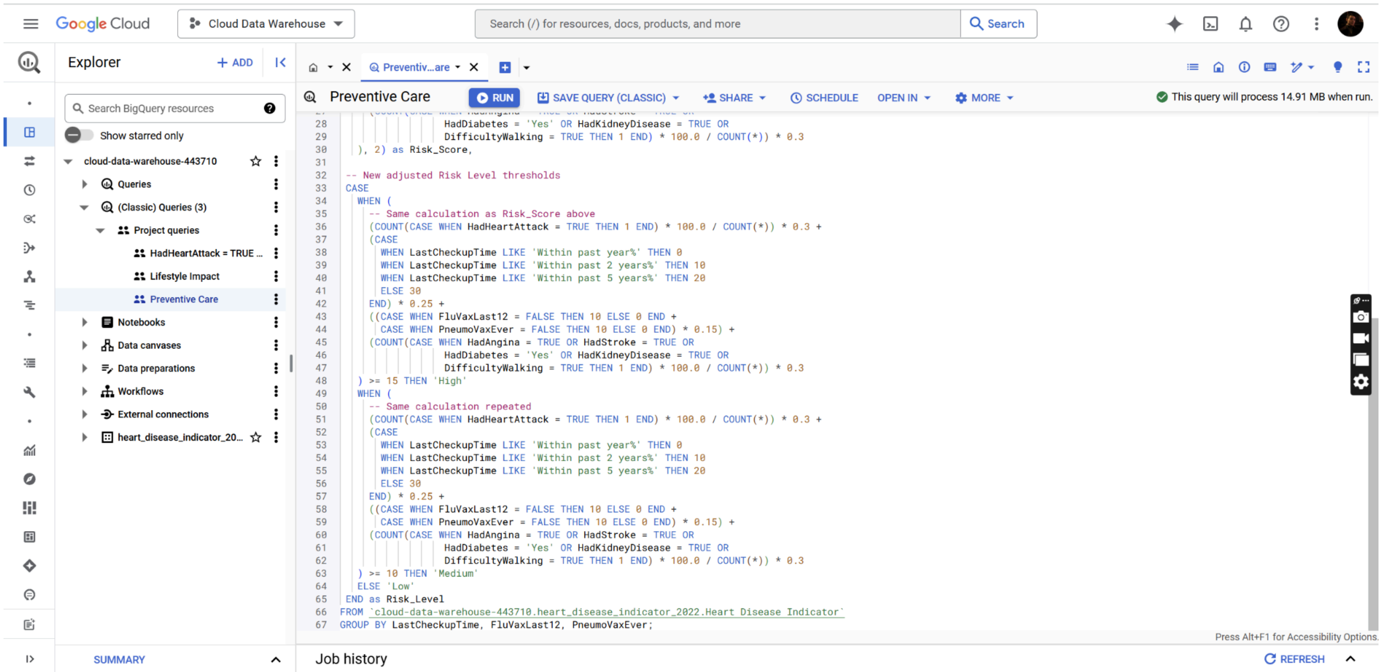
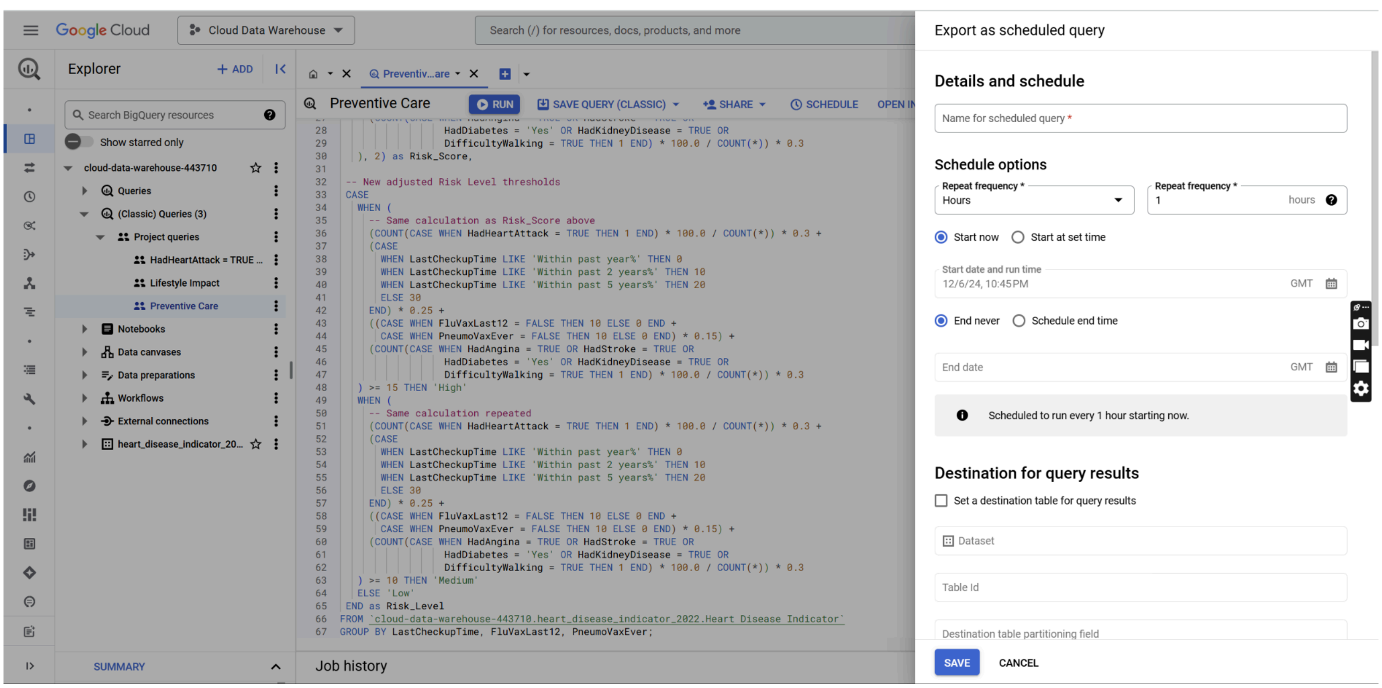
This query will process 14.91 MB when run.
SELECT
LastCheckupTime,
FluVaxLast12,
PneumoVaxEver,
COUNT(*) as Total_Count,
COUNT(CASE WHEN HadHeartAttack = TRUE THEN 1 END) as Heart_Attack_Count,
ROUND(COUNT(CASE WHEN HadHeartAttack = TRUE THEN 1 END) * 100.0 / COUNT(*), 2) as Heart_Attack_Rate,
-- Risk Score calculation from second version
ROUND(
(
-- Heart Attack Rate (30% weight)
(COUNT(CASE WHEN HadHeartAttack = TRUE THEN 1 END) * 100.0 / COUNT(*)) * 0.3 +
-- Checkup Recency Score (25% weight)
(CASE
WHEN LastCheckupTime LIKE 'Within past year%' THEN 0 -- Lower risk if recent checkup
WHEN LastCheckupTime LIKE 'Within past 2 years%' THEN 10
WHEN LastCheckupTime LIKE 'Within past 5 years%' THEN 20
ELSE 30 -- Higher risk if no recent checkup
END) * 0.25 +
-- Preventive Care Score (15% weight)
((CASE WHEN FluVaxLast12 = FALSE THEN 10 ELSE 0 END +
CASE WHEN PneumoVaxEver = FALSE THEN 10 ELSE 0 END) * 0.15) +
-- Health Conditions Score (30% weight)
(COUNT(CASE WHEN HadAngina = TRUE OR HadStroke = TRUE OR
HadDiabetes = 'Yes' OR HadKidneyDisease = TRUE OR
DifficultyWalking = TRUE THEN 1 END) * 100.0 / COUNT(*)) * 0.3
), 2) as Risk_Score,
-- New adjusted Risk Level thresholds
CASE
WHEN (
-- Same calculation as Risk_Score above
(COUNT(CASE WHEN HadHeartAttack = TRUE THEN 1 END) * 100.0 / COUNT(*)) * 0.3 +
(CASE
WHEN LastCheckupTime LIKE 'Within past year%' THEN 0
WHEN LastCheckupTime LIKE 'Within past 2 years%' THEN 10
WHEN LastCheckupTime LIKE 'Within past 5 years%' THEN 20
Press Alt+F1 for Accessibility Options.
Job history
REFRESH

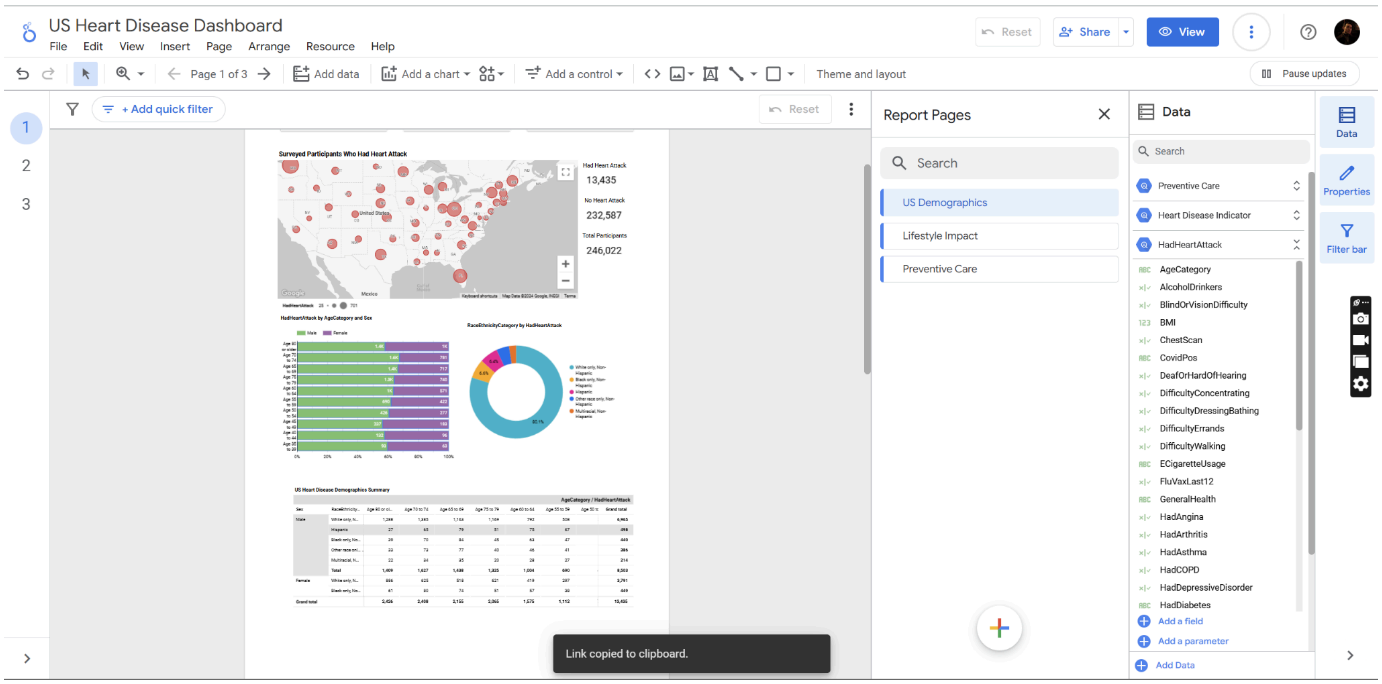


Query results were saved as tables for the subsequent data visualization step to use as data sources. BigQuery also allows scheduled query functionality, which helps update the data regularly when there is a large continuous amount of data ingestion.

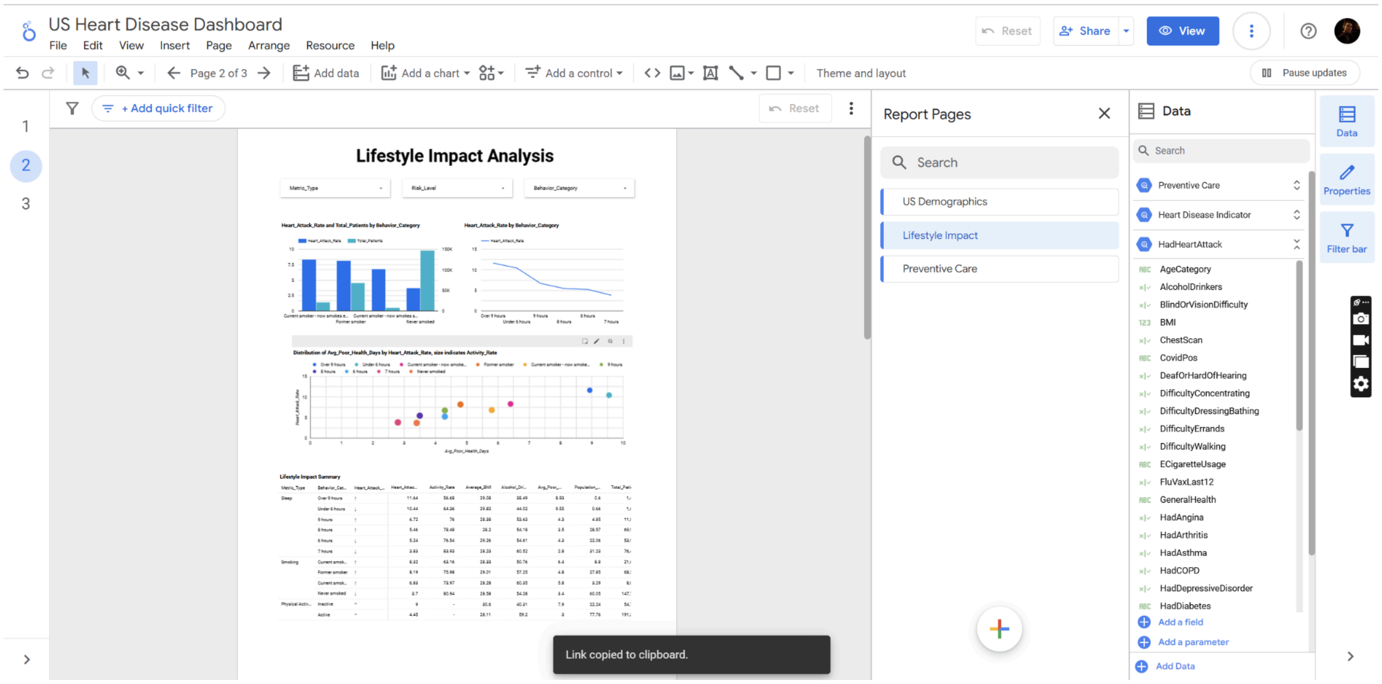


As for visualization implementation with Looker Studio, the dashboard design followed a user-centric approach with an emphasis on interactivity and accessibility. The intuitive interface is catered to both technical and non-technical users, using clear labeling, color-coded visualizations, and logical navigation. Filters for key variables are also included to enable in-depth analyses. The result is three dashboard types as follows:

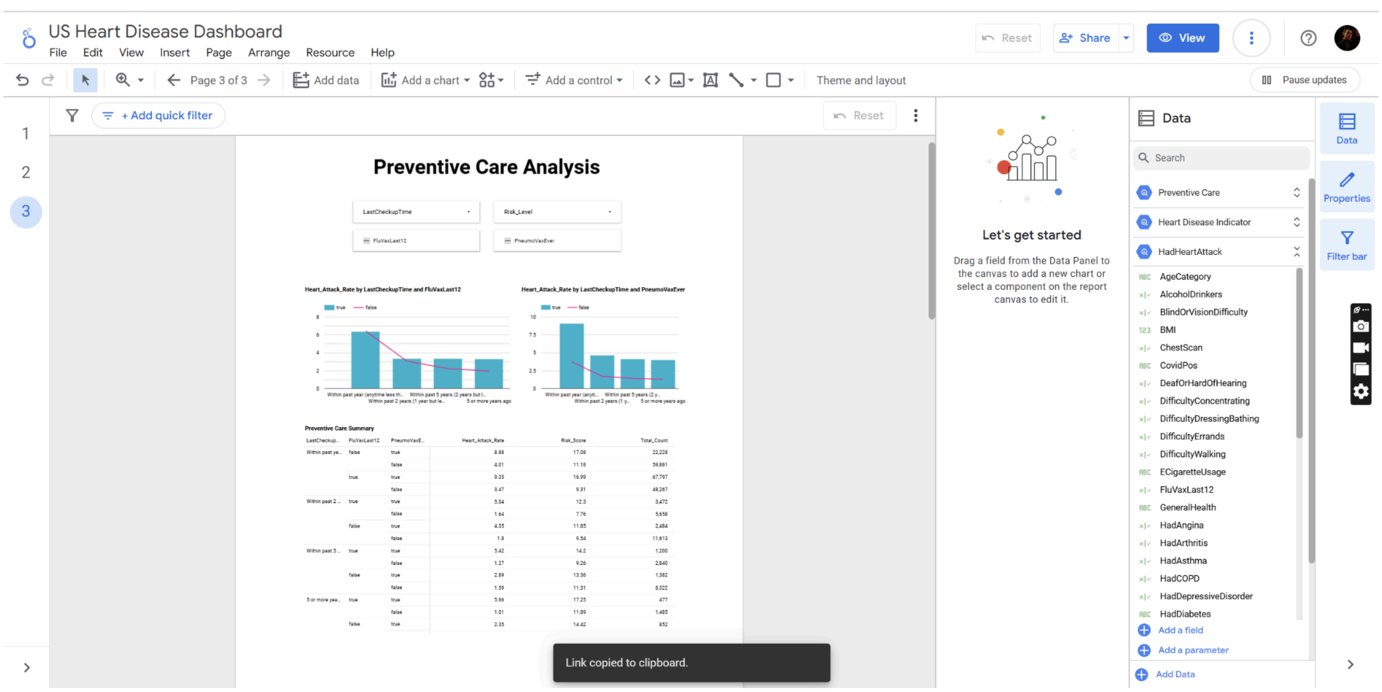
The demographic analysis dashboard provides an overview dot plot map of the geographic distribution of heart attack rates across U.S. states, with supporting charts on the demographic breakdown that allow public health administrators to identify at-risk populations. In this dashboard, public health administrators have access to dropdown filters, a bubble chart showing heart disease cases by region, a bar chart comparing heart attack rates by age and gender, a pie chart showcasing heart rate attack percentage by race, and a summary pivot table.



The lifestyle impact dashboard visualizes how behavioral factors correlate with cardiovascular outcomes. In this dashboard, medical researchers have access to dropdown filters, a scatter plot illustrating correlations between heart attack rates, physical activity, and sleep patterns, a pivot table, and two bar charts visualizing smoking and sleeping patterns in patients with heart disease history.



The preventive care dashboard tracks the effectiveness of intervention methods across population groups to help health providers evaluate and refine their prevention strategies. In this dashboard, healthcare providers have access to checkbox filters, dropdown filters, a pivot table, and bar charts comparing heart attack rates against vaccination types and checkup timelines.



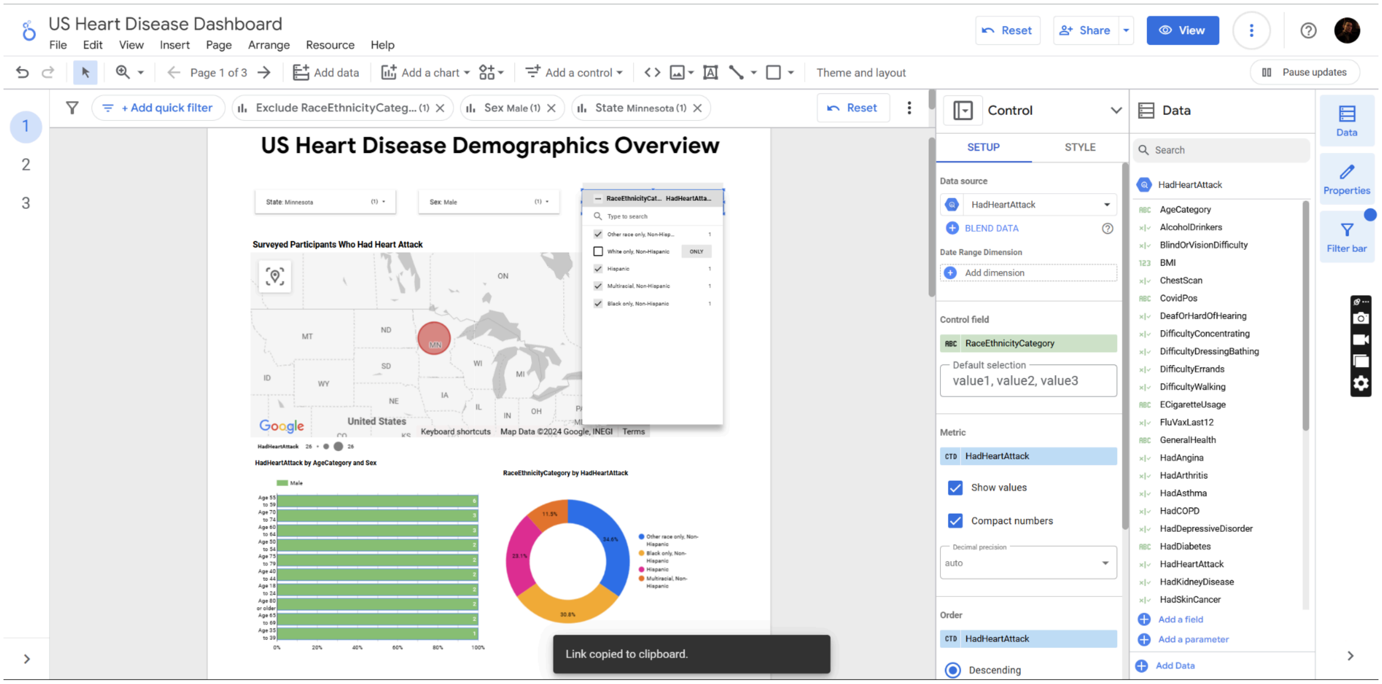
Suitability of Google Cloud BigQuery & Looker Studio

This section merely highlights the choice of Google BigQuery as the data warehousing solution, given that Looker Studio was chosen for obvious reasons - it is part of Google Cloud Platform's native services, allowing for smooth data pipeline integration. Compared to the Hadoop ecosystem like Cloudera HDP and similar data platforms like AWS Redshift, BigQuery can automatically allocate resources and scale to match large volumes of data and analytical demands without manual capacity planning, helping eliminate overheads. In contrast, Hadoop requires significant overheads in cluster management and configuration, despite its vast distributed computing power (StackShare, n.d.). In terms of query processing, BigQuery also outweighs the MapReduce approaches used in Hadoop and similar platforms. While MapReduce breaks queries into the map and reduce phases that need optimization or Cloudera HDP requires manual configuration of processing nodes and memory allocation, BigQuery's Dremel engine automatically optimizes queries using a columnar storage format (Tereshko & Tigani, 2016). This is much more efficient when analyzing specific health metrics; a notable example is when users examine the correlation between preventive care methods and heart attack rates across demographic groups.

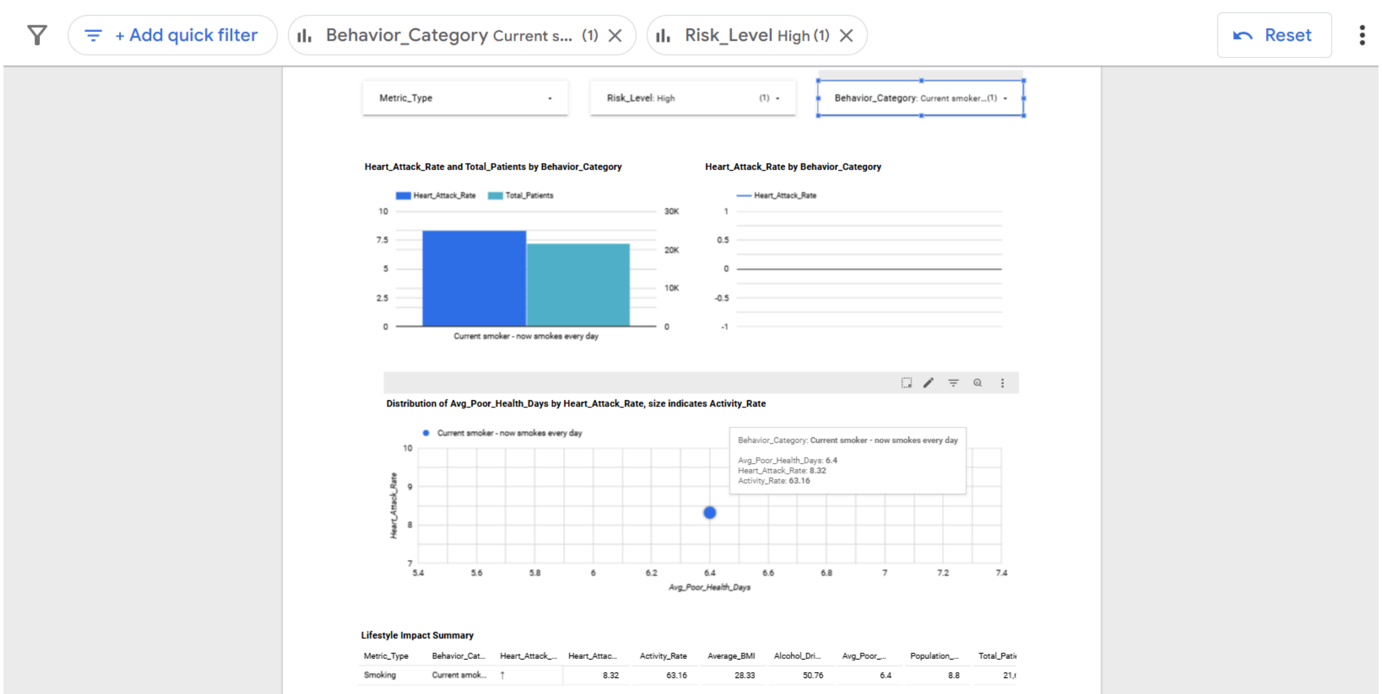
Software's Features

Dashboard Analytics and Visualization

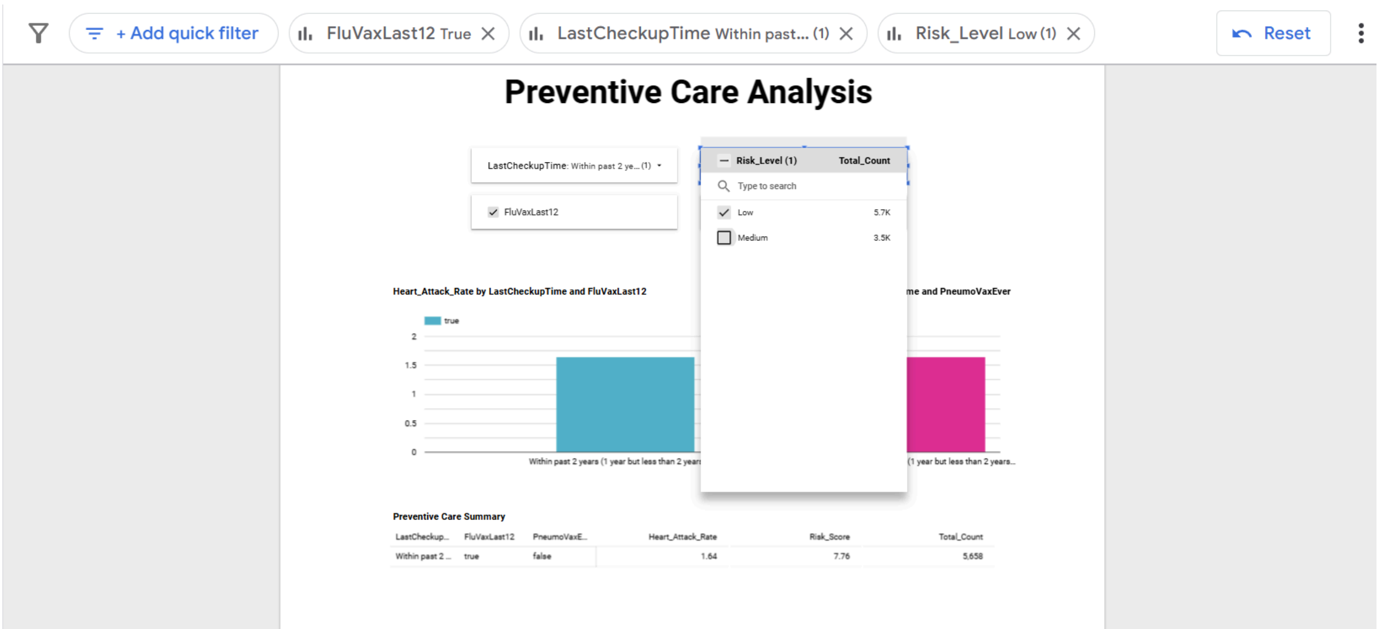
First, the demographic analysis dashboard enables public health administrators to visualize heart attack rates across regions through an interactive dot plot map interface. The dashboard supports real-time filtering by age groups, gender, and ethnicity, which can reveal hidden patterns among clustered data. For example, when viewing a certain region, users can immediately see how heart attack rates vary across different demographic segments and thus be able to identify population groups prone to heart disease. In addition, users can further deepen their analyses by clicking on specific demographic segments in the charts and the pivot table also enables cross-filtering to dynamically update data in all views. Please also refer to the Methodology and Implementation, where feature details of the dashboard have been discussed.



Second, the lifestyle impact dashboard helps medical researchers explore the correlational relationships between sleep patterns, physical activity levels, and smoking habits, as well as their combined impacts on heart attack risk. Filter and cross-filtering functionalities are also accessible in this dashboard to support researchers' in-depth analysis of how a certain behavioral category correlates to heart attack risk. Please also refer to the Methodology and Implementation, where similar feature details of the dashboard have been discussed.



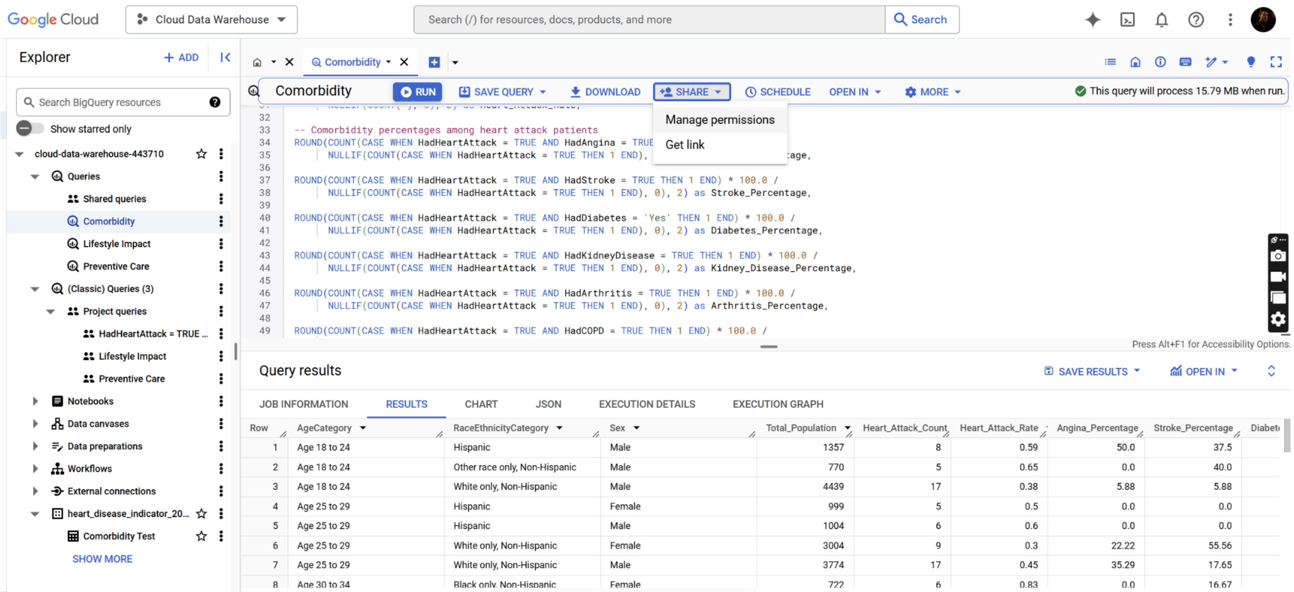
Third, the preventive care dashboard helps healthcare providers evaluate the impact of interventions. It tracks metrics such as regular checkup routines, vaccination rates, and their correlation with heart attack cases. The dashboard calculates risk scores for different preventive measures, allowing healthcare providers to tailor their strategies based on derived insights. Similarly, filter and cross-filtering functionalities also exist in this dashboard for in-depth analysis. Please also refer to the Methodology and Implementation, where similar feature details of the dashboard have been discussed.



Apart from the three current dashboards, users can modify dashboard layouts following their analytical needs and save their own versions. In addition, users can export results in different formats, from detailed CSV files to presentation PDF reports with relevant metadata and timestamp information.

Data Management and Processing

As mentioned in Methodology and Implementation, queries can be saved for later uses, shared with others for viewing purposes, and scheduled to update data in real-time. Also, while this dashboard solution does not utilize machine learning, BigQuery supports users in building predictive models, which is extremely insightful in determining the ordered importance of the existing risk factors concerning cardiovascular health. Additionally, as new survey data enters the platform, existing metrics and visualizations can be automatically updated in real-time without configurations, ensuring users always have access to the latest information.



Security

With Google Cloud Platform's role-based access control, administrators can define access permissions based on user roles so that necessary viewers/editors can access the data they have clearance for, while sensitive patient data's integrity and protection are secured. Furthermore, the platform maintains logs of all data access and analysis activities, which supports compliance requirements and security needs. Last but not least, all data, both at rest and in transit, undergoes encryption using industry-standard protocols, which is necessary given the sensitive nature of health information.

Worked Example

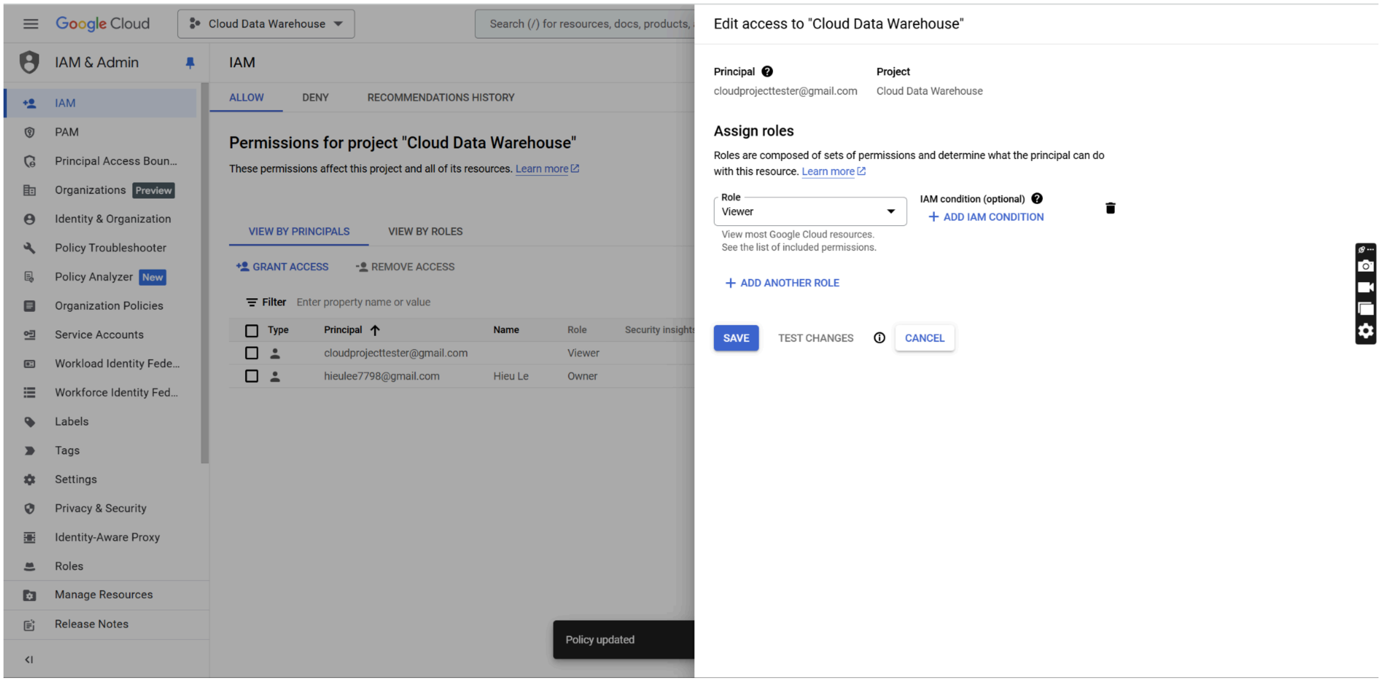
Necessary Credentials

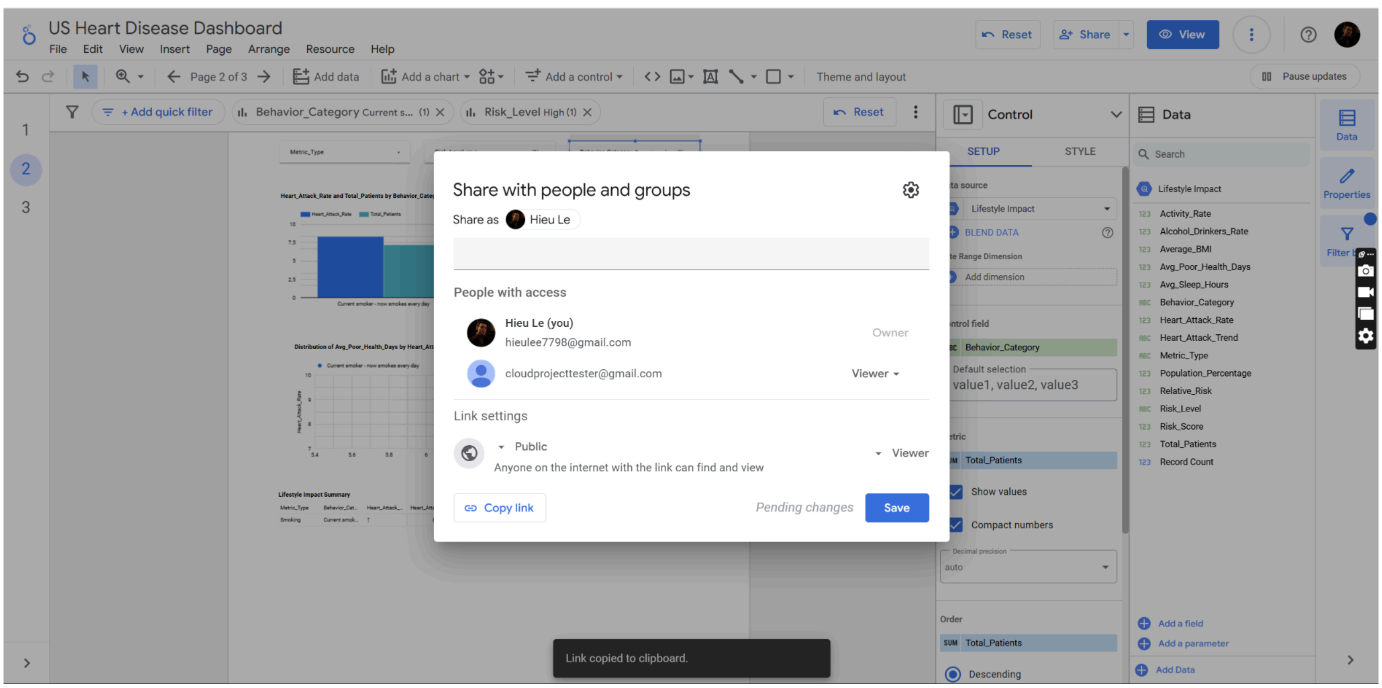
The following credentials were created for testing purposes.

Gmail: cloudprojecttester@gmail.com
Password: usheartdisease

Dashboard link (public view for anyone):
https://lookerstudio.google.com/reporting/5608abd6-c6c9-436b-965e-b742a8c57f44

BigQuery Project link (require the above credentials):
https://console.cloud.google.com/bigquery?invt=Abjcow&project=cloud-data-warehouse-443710&ws=!1m0





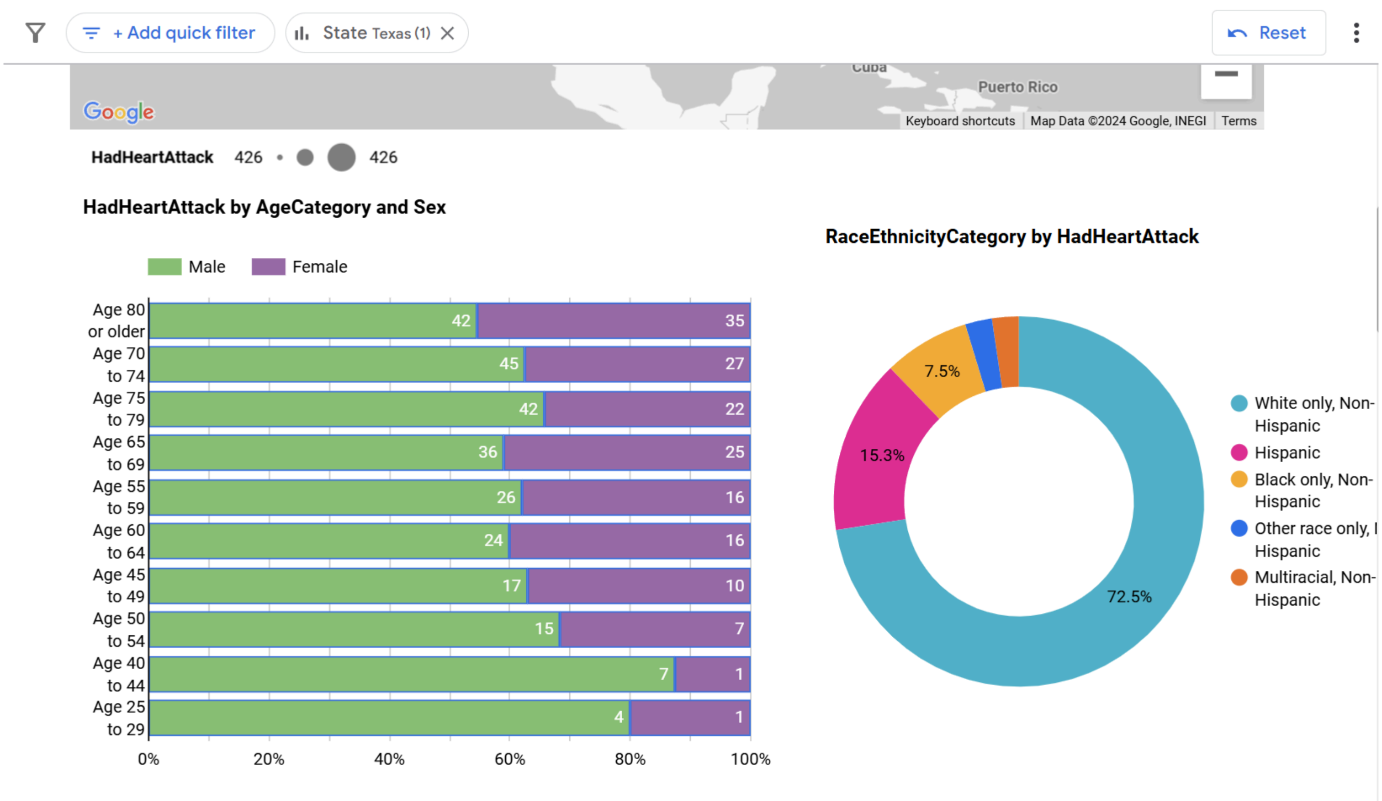
The account has been granted IAM access as a Viewer. One can use these credentials to access the Google Cloud BigQuery's project to view/run queries and see data tables. These can also access Looker Studio to use the dashboards, although the Link settings for the dashboards can be publicly viewed by anyone.

Scenario

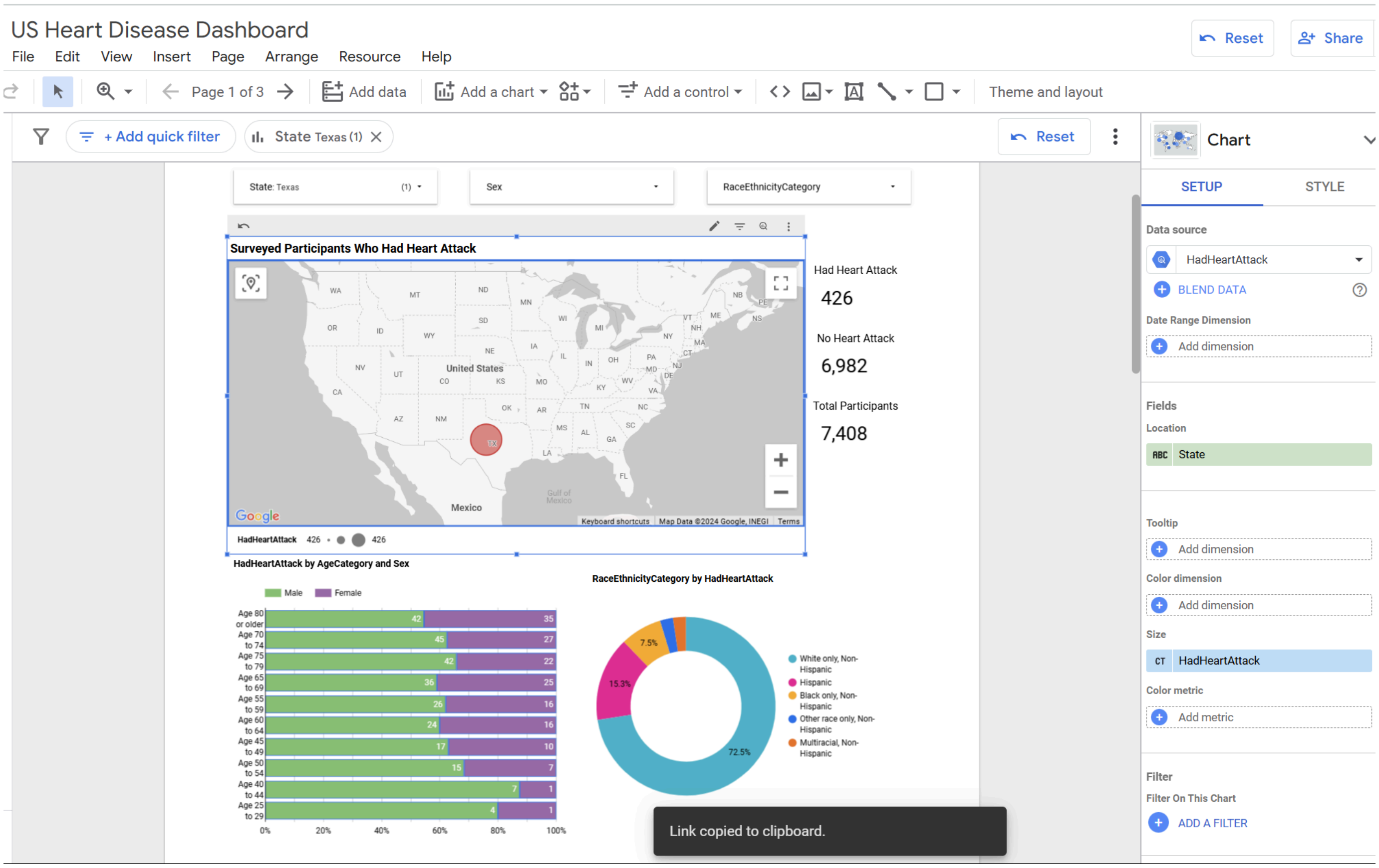
A public health official from the CDC works with a data analyst from their team to identify high-risk populations for heart disease. Their goal is to effectively allocate funding and design targeted health campaigns for these population groups. Given the existing data tables and dashboards, the official wants to uncover the U.S. states with the highest prevalence of heart disease and the demographic disparities in heart attack rates. Meanwhile, the data analyst seeks to leverage the unused data dimensions to explore comorbidity patterns with heart disease (e.g. HadSkinCancer, HadKidneyDisease, HadAsthma, etc.) associated with these populations.

User 1: Public health official

1. Access the dashboard:
   - The official opens the dashboard and begins on the Demographic Overview page.
2. Filter by State:
   - Using the state filter, the official selects a certain state like "Texas."
   - The dot plot map displays the dot representing Texas, while the bar chart, pie chart, and pivot table display a breakdown of heart attack rates by age and sex.
3. Insight:
   - The data reveals that white males aged 70–74 in Texas have the highest heart attack rates.

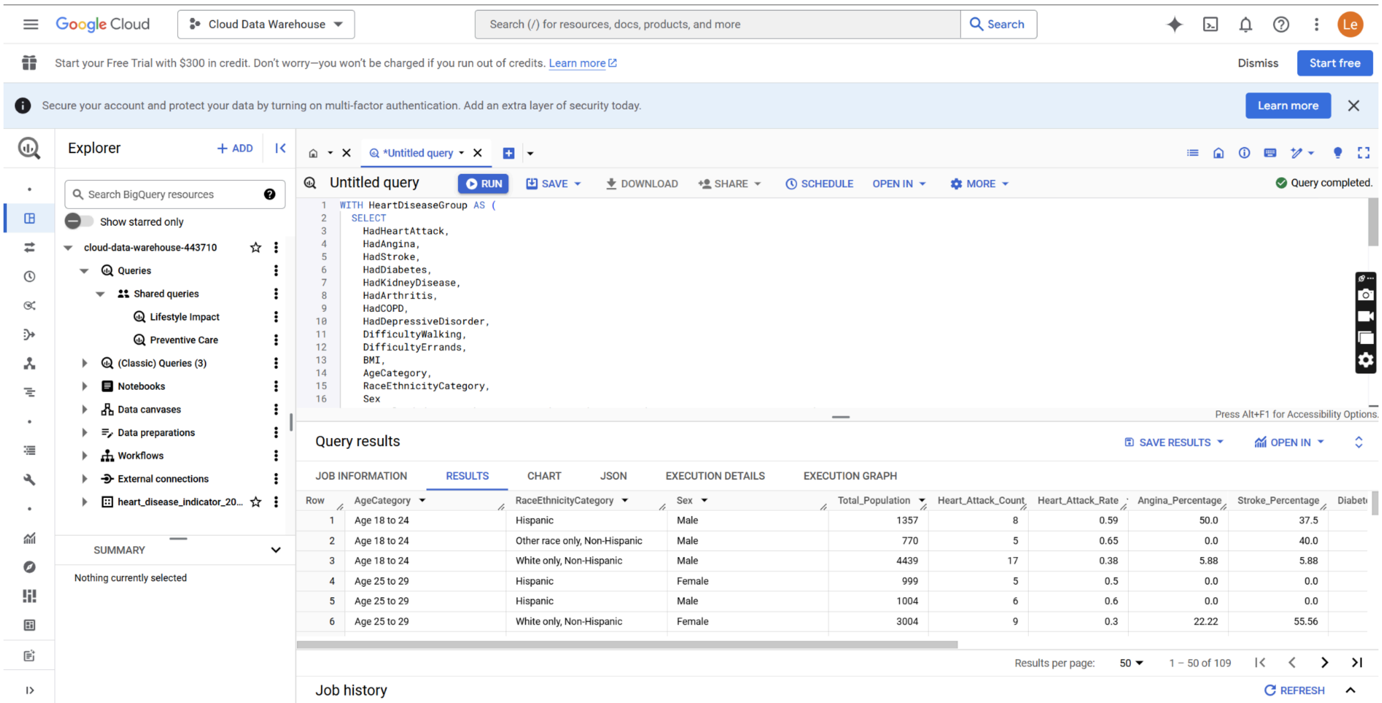


4. Generate reports:
   - The official exports visualizations and data summaries for internal presentations and funding allocation proposals.



User 2: Data analyst

1. Access [BigQuery Project]():
   ○ The analyst logs into the Google Cloud Console and navigates to the BigQuery project containing the heart disease dataset.
2. Compose a query:
   ○ The analyst uses SQL to query the dataset, focusing on heart disease and its relationship with comorbid conditions to uncover:
      ■ How heart disease risk and comorbidity patterns differ between men and women within the same age and racial/ethnic groups.
      ■ Whether certain combinations of demographic factors are associated with higher rates of specific comorbidities.
      ■ Where health disparities exist across multiple demographic dimensions.



3. Analyze query results:
    - The analyst reviews the query output, which includes:
        - Total patients by demographic segments (age/sex/race).
        - Heart attack rates as a percentage of total patients for different comorbidities.
4. Synthesizing results:
    - The analyst identifies demographic groups with the highest comorbidity rates for heart disease.
5. Actionable insights:
    - Based on the findings, the analyst makes recommendations for the target groups and reports to the public health official.

Conclusion

This combination tool of data warehouse, data analytics, and data visualization solutions has leveraged Google Cloud Platform's integrated services, aiding heart disease prevention and management. Its implementation utilizes BigQuery's serverless architecture, which showcases major advantages over traditional data platforms like Hadoop in terms of scalability, security, and operational efficiency. Combined with Looker Studio's seamless visualization capabilities, the tool achieves its objectives by enabling data-driven identification of cardiovascular risk patterns across demographics, supporting detailed analysis of lifestyle factors' impact on heart health, and providing healthcare providers with actionable insights for preventive care. The tool's success proves how cloud technologies can effectively close the gap between healthcare data collection and actionable insights and contribute to improved public healthcare management.

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